

sequence it has seen a million times—that could be accomplished with nothing more than memorizing sentences. What is impressive is that GPT-3 can be given a *novel* sequence that it has never seen before and still accurately predict the next word. This, too, clearly captures something that the human brain can _____.

Could you predict that the next word was *do*? I'm guessing you could, even though you had never seen that exact sentence before. The point is that both GPT-3 and the neocortical areas for language seem to be engaging in prediction. Both can generalize past experiences, apply them to new sentences, and guess what comes next.

GPT-3 and similar language models demonstrate how a web of neurons can reasonably capture the rules of grammar, syntax, and context if it is given sufficient time to learn. But while this shows that prediction is *part* of the mechanisms of language, does this mean that prediction is *all there is* to human language? Try to finish these four questions:

- If $3x + 1 = 3$, then x equals _____
- I am in my windowless basement, and I look toward the sky, and I see _____
- He threw the baseball 100 feet above my head, I reached my hand up to catch it, jumped, and _____
- I am driving as fast as I can to LA from New York. One hour after passing through Chicago, I finally _____

Here something different happens. In the first question, you likely paused and performed some mental arithmetic before being able to answer the question. In the other questions, you probably, even for only a split second, paused to visualize yourself in a basement looking upward, and realized what you would see is the ceiling. Or you visualized yourself trying to catch a baseball a hundred feet above your head. Or you imagined yourself one hour past Chicago and tried to find where you would be on a mental map of America. With these types of questions, more is happening in your brain than merely the automatic prediction of words.

We have, of course, already explored this phenomenon—it is simulating. In these questions, you are rendering an inner simulation, either of shifting values in a series of algebraic operations or of a three-

dimensional basement. And the answers to the questions are to be found only in the rules and structure of your inner simulated world.

I gave the same four questions to GPT-3; here are its responses (responses of GPT-3 are bolded and underlined):

- If $3x + 1 = 3$, then x equals **1**
- I am in my windowless basement, and I look toward the sky, and I see **a light, and I know that it is a star, and I am happy.**
- He threw the baseball 100 feet above my head, I reached my hand up to catch it, jumped, **and caught it. It was a lot of fun!**
- I am driving as fast as I can to LA from New York. One hour after passing through Chicago, I finally **get to see the Pacific Ocean.**

All four of these responses demonstrate that GPT-3, as of June 2022, lacked an understanding of even simple aspects of how the world works. If $3x + 1 = 3$, then x equals $\frac{2}{3}$, not 1. If you were in a basement and looked toward the sky, you would see your ceiling, not stars. If you tried to catch a ball 100 feet above your head, you would *not* catch the ball. If you were driving to LA from New York and you'd passed through Chicago one hour ago, you would not yet be at the coast. GPT-3's answers lacked common sense.

What I found was not surprising or novel; it is well known that modern AI systems, including these new supercharged language models, struggle with such questions. But that's the point: Even a model trained on the entire corpus of the internet, running up millions of dollars in server costs—requiring acres of computers on some unknown server farm—*still* struggles to answer commonsense questions, those presumably answerable by even a middle-school human.

Of course, reasoning about things by simulating also comes with problems. Suppose I asked you the following question:

Tom W. is meek and keeps to himself. He likes soft music and wears glasses. Which profession is Tom W. more likely to be?

- 1) *Librarian*
- 2) *Construction worker*

If you are like most people, you answered *librarian*. But this is wrong. Humans tend to ignore base rates—did you consider the *base number* of

construction workers compared to librarians? There are probably one hundred times more construction workers than librarians. And because of this, even if 95 percent of librarians are meek and only 5 percent of construction workers are meek, there still will be far more meek construction workers than meek librarians. Thus, if Tom is meek, he is still more likely to be a construction worker than a librarian.

The idea that the neocortex works by rendering an inner simulation and that this is how humans tend to reason about things explains why humans consistently get questions like this wrong. We *imagine* a meek person and compare that to an imagined librarian and an imagined construction worker. Who does the meek person seem more like? The librarian. Behavioral economists call this the representative heuristic. This is the origin of many forms of unconscious bias. If you heard a story of someone robbing your friend, you can't help but render an imagined scene of the robbery, and you can't help but fill in the robbers. What do the robbers look like to you? What are they wearing? What race are they? How old are they? This is a downside of reasoning by simulating—we fill in characters and scenes, often missing the true causal and statistical relationships between things.

It is with questions that require simulation where language in the human brain diverges from language in GPT-3. Math is a great example of this. The foundation of math begins with declarative labeling. You hold up two fingers or two stones or two sticks, engage in shared attention with a student, and label it *two*. You do the same thing with three of each and label it *three*. Just as with verbs (e.g., *running* and *sleeping*), in math we label operations (e.g., *add* and *subtract*). We can thereby construct sentences representing mathematical operations: *three add one*.

Humans don't learn math the way GPT-3 learns math. Indeed, humans don't learn *language* the way GPT-3 learns language. Children do not simply listen to endless sequences of words until they can predict what comes next. They are shown an object, engage in a hardwired nonverbal mechanism of shared attention, and then the object is given a name. The foundation of language learning is not sequence learning but the tethering of symbols to components of a child's already present inner simulation.

A human brain, but not GPT-3, can check the answers to mathematical operations using mental simulation. If you add one to three using your fingers, you notice that you always get the thing that was previously labeled *four*.